

Engineering Report:
Design and Analysis of the 805 Electric Race Vehicle
for the Waterloo EV Challenge



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Executive Summary

Following the cancellation of the 24 V race category, the previous Laurel Heights Secondary School (LHSS) Electric Vehicle Club race vehicle (803) is no longer suitable for the 12 V race format. The October 2025 race demonstrated that 803, equipped with a 48 V Saitta motor, could not sustain race-duration battery life at 12 V supply. A new vehicle, designated 805, has therefore been designed to address these limitations.

The 805 incorporates four major redesigns: a switch from steel to an aluminum frame, reducing frame mass from 64.4 kg to 22.5 kg; an Ackermann steering system in place of the previous anti-Ackermann geometry; two Kraken X60 brushless DC motors connected in parallel through a 9:1 gearbox and a 12:9 chain reduction (overall 12:1); and a new control system based on a Raspberry Pi 4B running CTR Electronics' Phoenix6 framework with TalonFX motor controllers over a CAN bus.

Quantitative analysis indicates that the 805 can operate continuously at 32.3 A on a single MTX-35 lead-acid battery for approximately 75 minutes, based on Peukert's Law with a Peukert constant of 1.125 and a Peukert capacity of 62.4 Ah. At cruise (43 km/h), the vehicle requires approximately 26.4 A combined motor current, leaving sufficient margin for transient loads during turn re-acceleration and pit-stop restarts. Acceleration from 0 to 43 km/h is projected to require approximately 16.4 s with a 60 A-per-motor burst current, while 27 to 43 km/h re-acceleration after each turn requires approximately 6.4 s.

Combining the lap-time and energy-budget analyses, the 805 is projected to complete approximately 40 laps of the 700 m course during the 70-minute race, with a 5-minute energy reserve. Final performance figures will be verified during physical testing prior to the October 2026 race.

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1.0 Introduction

1.1 Project Overview

The performance of the previous LHSS EVC race vehicle, 803, in the Waterloo EV Challenge October 2025 race was suboptimal. Although 803 led the race for the first 15 minutes, the battery capacity deteriorated rapidly, resulting in an inability to complete the race.

1.2 Problem Statement

The 803 vehicle was equipped with a single Saitta BR-119-R-68 motor, which is designed for 48 V operation. According to the manufacturer's technical specifications, this motor produces a torque of 12.1 Nm. However, due to the characteristics of a 48 V-rated motor, it does not operate efficiently under 12 V usage. The motor consumed amperage in excess of expectations, which contributed to 803's premature exit from the race. Additionally, the motor weighs 11 kg, increasing the overall mass of the vehicle and therefore the energy required for acceleration. Furthermore, 803 was equipped with an anti-Ackermann steering system, which is not advantageous for the relatively low-speed nature of this race.

1.3 Design Objectives

Following analysis of the previous race, the following design objectives were established:

1. Develop a new steering system that enables efficient turning;
2. Increase the vehicle's top speed to above 35 km/h;
3. Improve drivetrain efficiency such that the vehicle can complete the full race duration of 70 minutes;
4. Reduce the overall mass of the vehicle.

2.0 Background

2.1 Competition Overview

The competition is held in the parking lot adjacent to the East Campus of the University of Waterloo. The race track measures approximately 700 m in length and contains 3 major turns, as shown in Figure 1.

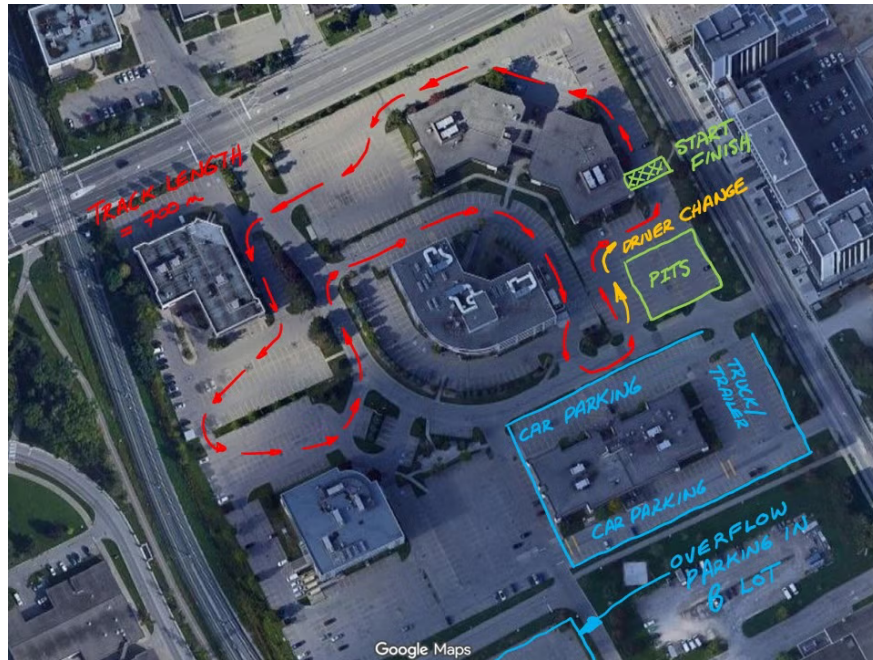


Figure 1: Race course layout for the May 2026 race.

2.2 Design Constraints

The rules governing the 2026 races are as follows:

1. The distance between tires must be at least 0.90 m;
2. Vehicle width must not exceed 1.2 m;
3. Maximum vehicle length is 3.5 m;
4. The driver must be in a seated, forward-facing position;
5. The frame must possess sufficient structural strength to protect the driver;
6. Two horizontal members spaced 6 to 10 inches apart must be installed to protect the driver's legs;
7. Permitted frame materials are steel and aluminum;
8. The roll bar must extend at least 50 mm above the top of the driver's helmet;
9. The steering system must enable a turn within a radius of 5 m;
10. Only the MTX-35 battery is permitted;

11. The battery must be securely mounted;
12. The driver must retain complete manual control of the vehicle.

3.0 Vehicle Design

3.1 Frame

3.1.1 Materials

Previous vehicle generations used steel for their frames. However, this material increased the overall mass of the vehicle. The current design therefore adopts aluminum as the primary frame material.

According to simulation results, the frame mass is reduced from 64.4 kg to 22.5 kg. This significantly reduces the energy required to accelerate 805 to its top speed.

3.1.2 Layout

The 805 vehicle shares the same frame architecture as 804. The design divides the vehicle into three sections: the leg, body, and back, as shown in Figure 2.

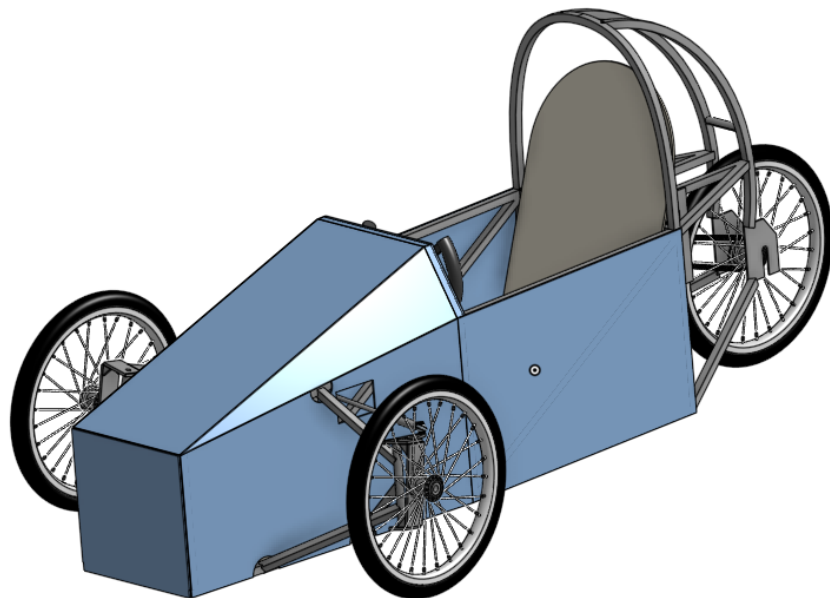


Figure 2: Overall design of the 805 vehicle.

Leg section. The leg section is the front portion of the vehicle. It accommodates the driver's legs, the steering system, and the MTX-35 battery.

The battery is mounted at the front of the vehicle. This location provides a more balanced weight distribution by counteracting the centre of mass of the driver. The battery is held in place by a battery holder consisting of two rods that secure the battery and four-sided barriers at the base, which prevent the battery from displacing during operation (Figure 3).

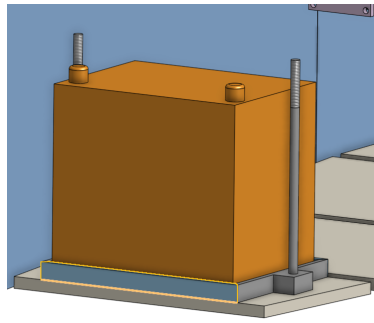


Figure 3: Battery mounting design.

A footrest is incorporated to prevent the driver from inadvertently dislodging the battery from its mounted position (Figure 4).

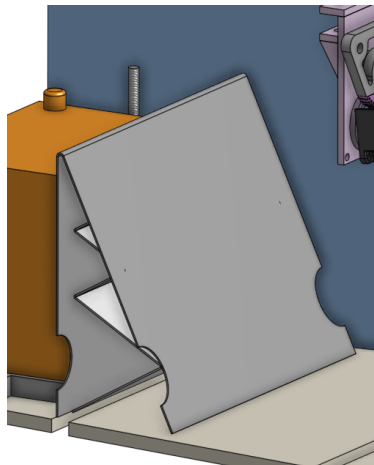


Figure 4: Footrest design.

The steering gearbox is bolted to the middle-top crossbar (Figure 5). Specific details regarding this gearbox are discussed in Section 3.2.

The leg section is enclosed by three panels (top, left, and right). These panels are intended to streamline the vehicle and reduce the drag coefficient (Figure 6).

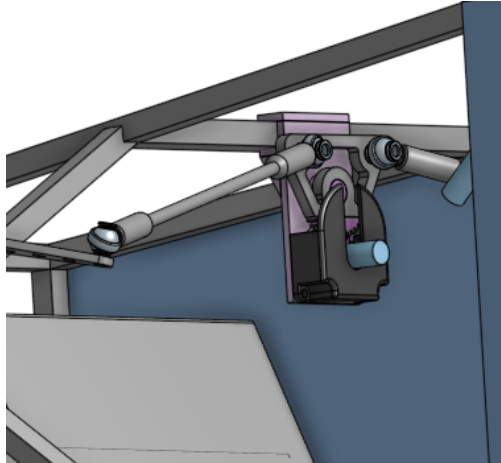


Figure 5: Steering gearbox location.

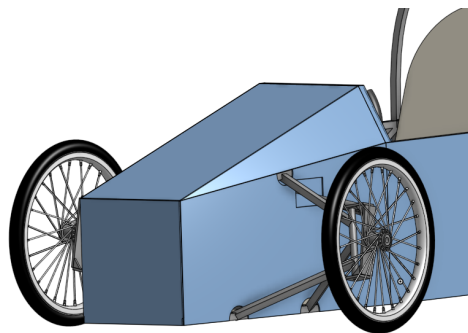


Figure 6: Front panel design.

Body section. The body section is where the driver sits and operates the vehicle. This section contains the kill switch (highlighted in yellow), the yoke, and the driver's seat (Figure 7).

The yoke is connected to three steering bars via two universal joints, which together control the steering gearbox. The driver sits on a wood plate inclined to both reduce the drag coefficient and enable the driver to maintain a forward visibility of at least 2 m. The kill switch is located on the right side of the driver to provide ready access in the event of an emergency.

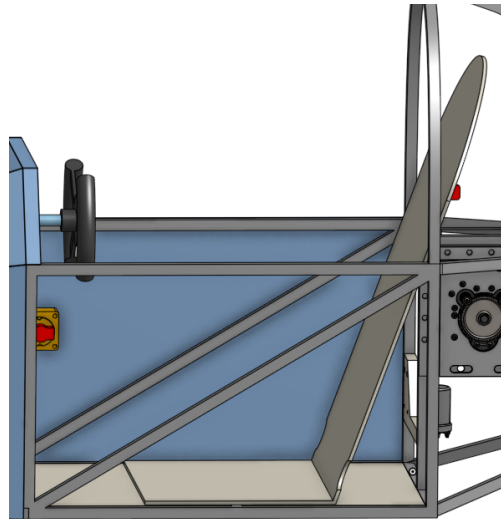


Figure 7: Body section design.

Back section. The back section of the vehicle houses all components of the drivetrain. Two Kraken X60 motors are mounted to a flipped gearbox, which is in turn attached to two mounting plates. After a gear reduction of 9:1, the gearbox transfers rotational energy to a sprocket connected to the rear wheel.

The two motors are connected via wiring to a CAN bus and a Raspberry Pi control unit. These components are mounted on the panel located above the rear wheel.

A roll cage extends from the bottom of the vehicle and arches over the driver's helmet, providing a clearance of 50 mm for driver safety. Two bent support beams connect the top of the roll cage to the rear mount of the vehicle, as shown in Figure 8.

The 805 is equipped with 19-inch diameter bicycle wheels at the two front positions and the rear position. This selection facilitates effective torque transfer from the motor to the ground. Furthermore, the bicycle wheel design significantly reduces frictional losses, allowing the vehicle to operate efficiently.

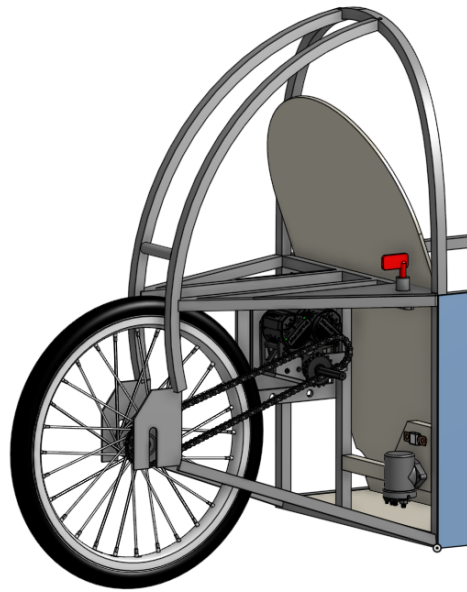


Figure 8: Back section design.

3.2 Steering System

[Section to be completed: detailed description of the Ackermann steering geometry, gearbox specifications, linkage dimensions, and turning radius calculation.]

3.3 Drivetrain

3.3.1 Overview

In light of the unsatisfactory results of the October race, the team concluded that 803 lacked the efficiency and energy required to complete the race. Accordingly, a motor designed for 12 V operation was selected for the 805 vehicle.

After preliminary research, the design team identified several industrial motors rated for 12 V operation. However, none of the corresponding manufacturers responded to inquiries, and this option was therefore not pursued.

Robotics motors became the only viable option for the team. After consultation with several FIRST robotics team members within the club, the Kraken X60 was identified as the optimal motor for this application [1]. Other motors that were considered include the Kraken X44, the NEO Vortex, and the NEO 2.0.

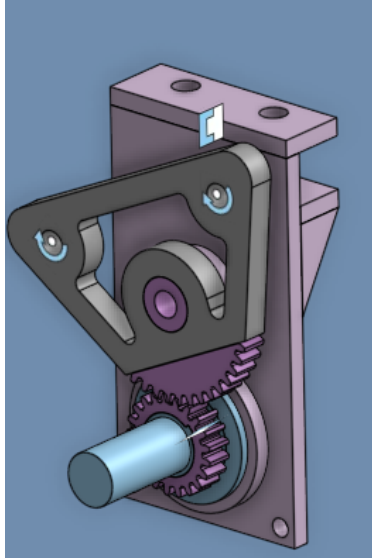


Figure 9: Steering system overview.

3.3.2 Performance Data

Using data acquired from publicly available sources, the performance graph shown in Figure 10 was generated to determine the operating RPM, torque, and current of the Kraken X60.

The horizontal axis represents the motor RPM. The blue curve indicates the output power (W); the purple curve indicates the current draw (A); the green curve indicates the torque (Nm); and the red curve indicates the efficiency. The interactive version of the graph is available at [2].

3.4 Electrical Architecture

3.4.1 Phoenix6 Software Framework

In order to control the two Kraken X60 motors, the Phoenix6 software framework by CTR Electronics is used to interface with the TalonFX motor controllers built into each motor [3]. The RoboRIO, an FRC-specific robotics controller, was not selected due to its cost. A Raspberry Pi 4B was chosen instead to run the Phoenix6 framework.

The architecture of the entire drivetrain is shown in Figure 11.

The system includes the following components:

1. **Throttle:** input device that allows the driver to control the power delivered to the motors;
2. **Analog-to-Digital converter:** a custom circuit that accepts a 0.8 to 4.2 V signal and outputs UART serial data to the Raspberry Pi;

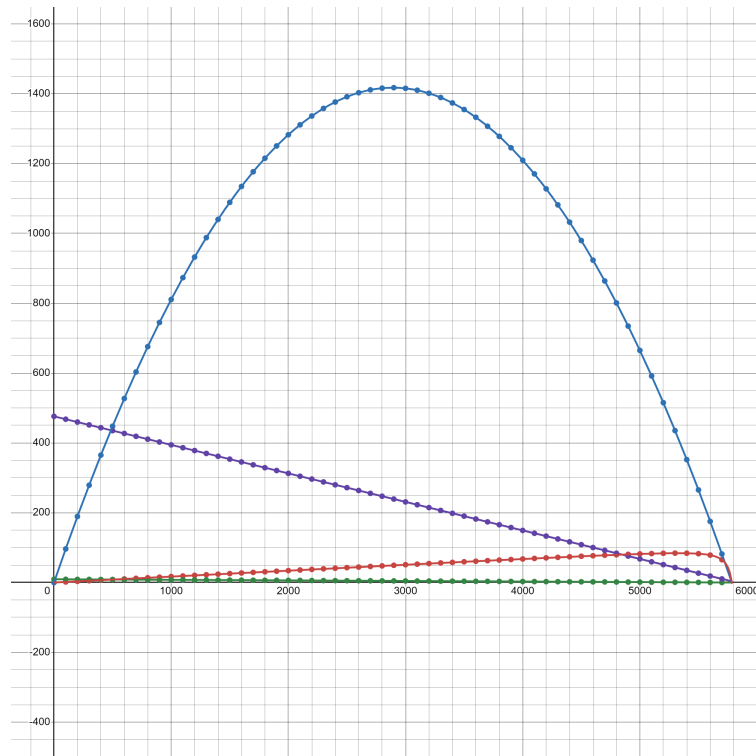


Figure 10: Kraken X60 performance characteristics.

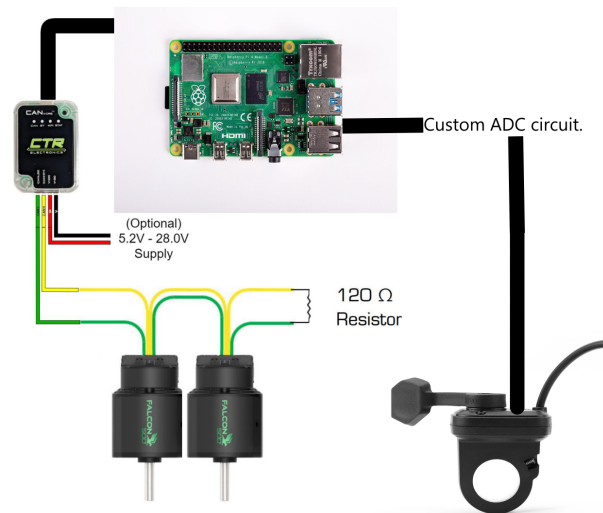


Figure 11: Electric drivetrain architecture.

3. **Raspberry Pi 4B:** processes throttle data and transmits CAN frames over the CAN bus to control the motors;
4. **Canivore:** a USB-to-CAN adapter manufactured by CTR Electronics;
5. **Kraken X60:** two motors connected in tandem to drive the rear wheel of the vehicle.

4.0 Drivetrain Analysis

4.1 Maximum Continuous Current

A preliminary calculation is required to determine the maximum continuous current that can be delivered to the motors while ensuring that the vehicle can complete the race.

The 12 V race format requires 805 to use a single MTX-35 battery. The MTX-35 is an absorbent glass mat (AGM) lead-acid battery with the specifications listed in Table 1 [4].

Table 1: MTX-35 battery specifications.

Parameter	Value
Mass	18.1 kg
Reserve capacity at 25 A	100 minutes (1.67 h)
Nominal voltage	12 V
Capacity (20-hour rate)	55 Ah
Maximum current rating	800 A
Chemistry	AGM lead-acid

Lead-acid battery behaviour is more accurately represented by Peukert's Law than by Watt's Law [5]:

$$C_p = I^k \times t \quad (1)$$

where C_p is the Peukert capacity (the equivalent capacity at a one-ampere discharge rate, expressed in ampere-hours); I is the actual discharge current (in amperes) drawn by the load; t is the actual discharge time (in hours); and k is the Peukert constant (dimensionless).

The Peukert constant k is determined for the MTX-35 from Equation 2:

$$k = \frac{\log(t_2) - \log(t_1)}{\log(I_1) - \log(I_2)} \quad (2)$$

Two discharge data points are obtained from the manufacturer's 20-hour rate and reserve capacity rate. The 20-hour rate corresponds to the rated capacity of 55 Ah, yielding a discharge

current of 2.75 A. The reserve capacity rate gives a discharge time of 1.67 h at 25 A. Substituting these values into Equation 2:

$$k = \frac{\log(1.67) - \log(20)}{\log(2.75) - \log(25)} = 1.125$$

Using the calculated Peukert constant in Equation 1, the maximum continuous current that can be drawn over the race is determined. The race duration is taken to be 75 minutes (1.25 h), which provides a 5-minute reserve over the 70-minute target:

$$2.75^{1.125} \times 20 = I_{race}^{1.125} \times 1.25$$

$$I_{race} = 32.3 \text{ A}, \quad C_p = 62.4 \text{ Ah}$$

The 805 vehicle may therefore draw a maximum continuous current of 32.3 A, with a Peukert capacity of 62.4 Ah.

4.2 Motor Performance at Maximum Current

Because the 805 employs two Kraken X60 motors connected in parallel through a single flipped gearbox, the current per motor is approximately 16.2 A continuous (half of the system current).

From the performance graph in Figure 10, each Kraken X60 operates at the following point at 16.2 A:

- Power: 158.8 W;
- Efficiency: 76.7%;
- RPM: 5624 RPM;
- Torque: 0.258 Nm.

For the parallel-connected pair, the combined performance is therefore 317.6 W of output power, 0.516 Nm of torque, and 5624 RPM at the motor shaft.

The selected gearbox provides a 9:1 reduction (motor to gearbox output). Its output is then coupled to the rear wheel through a chain drive with a 12:9 reduction (driving sprocket: 9 teeth on the gearbox output shaft; driven sprocket: 12 teeth on the wheel). The overall reduction from motor to wheel is therefore $9 \times \frac{12}{9} = 12:1$.

Applying the inverse relationship between torque and rotational speed across a gear reduction:

$$\frac{\tau_{input}}{\tau_{output}} = \frac{\text{Gear input}}{\text{Gear output}} \quad (3)$$

$$\begin{aligned}\tau_{output} &= 0.516 \times 12 = 6.192 \text{ Nm} \\ \frac{RPM_{input}}{RPM_{output}} &= \frac{\text{Gear output}}{\text{Gear input}} \\ RPM_{output} &= \frac{5624}{12} = 468.7 \text{ RPM}\end{aligned}\tag{4}$$

The wheel therefore operates at 468.7 RPM with an output torque of 6.192 Nm.

4.3 Stages of the Race

The race is run primarily at constant speed; however, turns and starts affect the current drawn from the battery. Accordingly, the 805 must be analyzed under the following three operating conditions:

1. Cruise;
2. Acceleration: 0–43 km/h (race start and after each pit stop);
3. Acceleration: 27–43 km/h (recovery from each major turn).

The current required to maintain a top speed of 43 km/h is determined using Newton's second law applied to the vehicle:

$$F_{net} = F_A - F_f - F_{AR}\tag{5}$$

where $F_{net} = ma$ is the net force on the vehicle; F_A is the propulsive force provided by the Kraken motors; F_f is the rolling resistance force opposing motion; and F_{AR} is the aerodynamic drag force opposing motion.

4.3.1 Cruise

At cruise, $F_{net} = 0$, hence:

$$F_A = \mu mg + \frac{1}{2}\rho v^2 C_D A$$

The parameters and assumptions used in this calculation are listed in Table 2.

$$F_A = (0.005)(116.5)(9.81) + \frac{1}{2}(1.225)(11.94)^2(0.35)(0.42) = 18.6 \text{ N}$$

The required propulsive force is therefore 18.6 N. Applying the relationship between force at the wheel and torque:

$$\tau = F_A \cdot r_{wheel} = 18.6 \times 0.2413 = 4.48 \text{ Nm}$$

Table 2: Cruise parameters and assumptions.

Symbol	Description	Value
μ	Rolling resistance coefficient (bicycle on asphalt)	0.005
m	Total vehicle mass	116.5 kg
g	Gravitational acceleration	9.81 m/s ²
ρ	Air density	1.225 kg/m ³
v	Velocity (43 km/h)	11.94 m/s
C_D	Drag coefficient	0.35
A	Frontal cross-sectional area	0.42 m ²

After the 12:1 gear reduction and the parallel split between the two motors, the torque required from each individual motor is approximately 0.187 Nm. The corresponding operating point read from the performance graph (Figure 10) is summarized in Table 3.

Table 3: Cruise operating point per motor at 43 km/h.

Quantity	Value
Mechanical power per motor	111.0 W
Efficiency	~70%
Current per motor	13.2 A
Combined cruise current	26.4 A
Motor RPM	5680 RPM

The combined cruise current of 26.4 A is well within the 32.3 A continuous budget established in Section 4.1, leaving a margin for transient loads.

4.3.2 Acceleration: 0–43 km/h

This phase applies at the start of the race and after each pit stop, and therefore occurs three times during the race. From a full stop, the 805 must accelerate to its cruise speed of 43 km/h. Applying Equation 5 with $F_{net} = m(dv/dt)$:

$$m \frac{dv}{dt} = F_A(I) - \mu mg - \frac{1}{2} \rho v^2 C_D A$$

The continuous current limit of 16.2 A per motor is insufficient for practical acceleration: at this current the propulsive force is only 25.7 N, giving an initial acceleration of 0.17 m/s² and a 0–43 km/h time exceeding 90 seconds. The TalonFX motor controllers therefore permit a short-duration current burst of approximately 60 A per motor (120 A combined) during these acceleration phases, which is well within the 800 A maximum rating of the MTX-35.

At a burst current of 60 A per motor:

- Torque per motor: $K_t \cdot I \approx 0.96 \text{ Nm}$ (using $K_t \approx 0.0159 \text{ Nm/A}$);
- Combined wheel torque after 12:1 reduction: $\sim 23 \text{ Nm}$;
- Propulsive force at the wheel: $F_A \approx 95 \text{ N}$;
- Initial acceleration: $a(0) \approx 0.77 \text{ m/s}^2$.

Solving the resulting first-order ODE numerically yields a 0 to 43 km/h time of approximately 16.4 s, a distance of approximately 100 m, and an electrical energy consumption of approximately 6.5 Wh per start. Across three pit-related starts, the total energy consumed during this phase is approximately 19.5 Wh.

4.3.3 Acceleration: 27–43 km/h

This phase models the re-acceleration following each major turn. The cornering speed is taken to be 27 km/h (7.50 m/s), which respects the 5 m turning radius constraint and provides safe lateral acceleration on asphalt.

Using the same 60 A per motor burst current and the same governing equation, the re-acceleration analysis yields a 27 to 43 km/h time of approximately 6.4 s, a distance of approximately 63 m, and an electrical energy consumption of approximately 2.5 Wh per turn exit.

The course contains 3 major turns per lap. The combined cruise and turn-burst energy demand is the dominant factor in the lap-count estimate presented in Section 5.

5.0 Results

The 805 is currently in late-stage development and will not be available to participate in the May 2026 race. The vehicle is instead targeted for the October 2026 race. Based on present calculations, the projected performance characteristics are summarized in Table 4.

5.1 Lap-Count Estimation

The total number of laps is estimated by combining the lap-time analysis with the energy budget over the 70-minute race.

Lap time. The 700 m course is divided into a cruise segment of approximately 271 m (where the vehicle holds 43 km/h) and three turn zones of roughly 143 m each (deceleration, cornering at 27 km/h, and re-acceleration). The estimated lap time is approximately 69 s.

Table 4: Projected performance characteristics of the 805.

Parameter	Projected Value
Top speed (cruise)	43 km/h
Cornering speed	27 km/h
Acceleration time, 0–43 km/h	~16.4 s
Acceleration time, 27–43 km/h	~6.4 s
Maximum continuous current	32.3 A
Cruise current (combined)	26.4 A
Race duration target	70 min (with 5-min reserve)
Number of pit stops	3 (driver changes)
Turning radius	<i>[to be measured]</i>

Pit stops. The race plan includes 3 pit stops for driver changes. Each stop is estimated at approximately 30 s of stationary time, plus a small additional time penalty due to 0–43 km/h re-acceleration relative to cruise. The combined pit cost is approximately 90–115 s over the race.

Time-limited lap count. Subtracting pit time from the 4200 s race duration and dividing by lap time yields an upper bound of approximately 59 laps.

Energy-limited lap count. Per-lap energy consumption is approximately 10 Wh (dominated by the three turn re-accelerations), and each pit-stop restart adds approximately 6.5 Wh. Against the 484.5 Wh battery budget, this gives an upper bound of approximately 46 laps.

Conclusion. Energy is therefore the binding constraint. After applying a margin of approximately 10–15% for real-world losses (weather, manufacturing tolerances, voltage sag, and additional drag), the 805 is projected to complete approximately **40 laps** during the 70-minute race.

These figures are projections only. Actual values are subject to weather conditions, additional friction, drag, voltage sag, and manufacturing-related energy losses. Final results will be reported following physical testing of the completed vehicle.

6.0 Conclusions

In order to remain competitive in the upcoming race, LHSS EVC has redesigned the steering, electrical, and motor systems of the vehicle, as well as the frame material, to maximize efficiency and motor power output.

The steering system has been changed from the anti-Ackermann configuration used on 804 to an Ackermann configuration. This system is expected to provide greater stability during turns

and reduce tire slip, thereby reducing drag in turns and allowing higher re-acceleration speeds following each major turn.

The 805 incorporates two Kraken X60 motors, which operate natively at 12 V. This configuration enables more efficient energy conversion. Because the Kraken motors operate at high RPM, a reducing gearbox combined with a chain drive (overall 12:1) is used to produce the torque and rotational speed required at the wheel.

The new motor system also required a new wiring architecture. A CAN bus control unit and a Raspberry Pi 4B running the Phoenix6 framework were incorporated to program and command the motor controllers for optimal efficiency.

The 805's frame material has been changed to aluminum, which has a lower density and therefore lower mass than steel. This reduces the energy required to accelerate and decelerate the vehicle, and the energy savings contribute to higher acceleration and a higher top speed.

Quantitative analysis projects a top speed of 43 km/h, a 0–43 km/h acceleration time of approximately 16 s under burst current, and approximately 40 completed laps during the 70-minute race.

7.0 Recommendations

The following actions are recommended to validate and refine the 805 design:

1. **Complete physical assembly** of the 805 vehicle in time for on-track testing prior to the October 2026 race.
2. **Tune the burst current limits** in the TalonFX firmware (via Phoenix6) to balance acceleration time against energy consumption over the full race.
3. **Implement telemetry logging** (motor current, voltage, RPM, battery state of charge) so that race data can be used to refine the model for the next vehicle generation.

References

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